

# Fried's identity at a crossing metric: the intuitive view

How a resonance walks to zero, drawn

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`fried_counterexample_of_resolvent_inputs`

a conditional reduction from cited inputs; not yet verified by a mathematician other than the author

## Abstract

Fried's conjecture identifies a dynamical number (the Ruelle zeta function at  $s = 0$ ) with a topological one (Reidemeister torsion). This document draws a construction in which the identity, as usually stated, fails: a closed hyperbolic 3-manifold with  $b_1 = 1$ , a unitary character  $\chi_\theta$  close to trivial, and a conformal deformation of the hyperbolic metric. Two dials are turned. The first ( $\theta$ ) plants a real resonance a distance  $\Theta(\theta^2)$  from zero; the second (the metric) walks it to zero. At the crossing, a pole and a zero of  $\zeta$  arrive together but at different speeds, and the value  $\zeta(0)$  jumps by a factor  $\Theta(\theta^2)$ . Chaubet–Dang's refined identity absorbs the jump; the naive identity does not. What is machine-checked is the reduction: granting six groups of facts cited from the literature about the two resonance clusters near 0, the value at the crossing is  $\tau_R \cdot (\text{pole rate})/(\text{zero rate}) \neq \tau_R$ . That the geodesic flow has those properties is taken from the cited papers and has not been verified by a mathematician other than the author.

## 1 What Fried says, and the two torsions

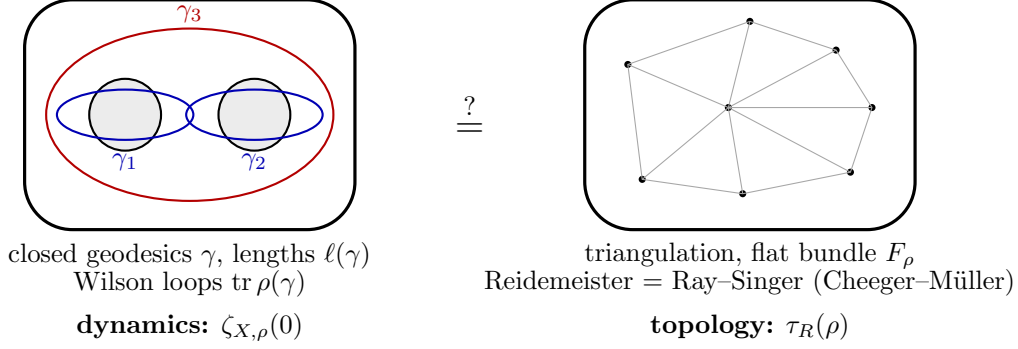
Let  $Z$  be a closed negatively curved manifold,  $\mathcal{S} = S^*Z$  its unit cotangent bundle,  $X$  the geodesic vector field, and  $\rho$  a unitary representation of  $\pi_1(Z)$ . The Ruelle zeta function is a product over primitive closed geodesics  $\gamma$ ,

$$\zeta_{X,\rho}(s) = \prod_{\gamma} \det(1 - \varepsilon_{\gamma} \rho(\gamma) e^{-s\ell(\gamma)}), \quad \varepsilon_{\gamma} = \text{sign} \det(1 - P_{\gamma}),$$

so that  $\log \zeta$  is a sum of Wilson loops  $\text{tr} \rho(\gamma^j)$  weighted by  $e^{-sj\ell(\gamma)}$ . It continues meromorphically to  $\mathbb{C}$ ; its zeros and poles are the Pollicott–Ruelle resonances of  $L_X$  acting on  $E$ -valued forms, counted with the sign  $(-1)^k$  of the form degree. On the topological side, if  $\rho$  is acyclic ( $H^{\bullet}(Z; \rho) = 0$ ) the Reidemeister torsion  $\tau_R(\rho)$  is a combinatorial invariant of a triangulation, equal by Cheeger–Müller to the analytic (Ray–Singer) torsion. Fried's conjecture, in the form stated by Dang–Guillarmou–Rivière–Shen (DGRS) for acyclic unitary  $\rho$ , is the naive identity

$$|\zeta_{X,\rho}(0)|^{(-1)^q} = \tau_R(\rho), \quad q = \dim E_s,$$

proved by Fried for hyperbolic metrics and by DGRS for small perturbations of them. Here  $q = 2$ , so the exponent is  $+1$ .



**Figure 1:** The identity to test. Left: the dynamical side counts closed geodesics, each weighted by the trace of its holonomy. Right: the topological side is a combinatorial determinant, which by Cheeger–Müller equals a spectral determinant of Laplacians. Fried asks whether the two agree at  $s = 0$ . On the metrics drawn in §5–§7 they do not.

The plan is to keep  $\rho$  acyclic (so  $\tau_R$  is defined and  $\zeta(0)$  is finite and nonzero at the hyperbolic metric) and to move the metric until a resonance of  $L_X$  sits exactly at  $s = 0$ . The order of  $\zeta$  there is  $\sum_k (-1)^k c_k$ , where  $c_k$  is the number of degree- $k$  resonant states at 0; the value of  $\zeta(0)$  is a separate question, and it is the value that fails.

**Vocabulary shared with the Lean statement.** Near  $s = 0$  the resonances of interest form two small clusters: a rank-2 cluster in degree 1, encoded by a  $2 \times 2$  matrix  $M(\theta, \tau)$ , and a rank-4 cluster in degree 2, encoded by a  $4 \times 4$  matrix  $A(\theta, \tau)$ . A *branch* is one eigenvalue of a cluster matrix followed continuously in  $(\theta, \tau)$ . Degree 1 has the *closed branch* and the *twin*. Degree 2 has the two *locked branches* (the eigenvalues of  $M$ , lifted by  $d_0$ ) and two *pure branches*, the roots of  $Q = X^2 - e_1 X + e_2$ ; the pure branch nearest 0 is written  $z$ . The six figures of §3–§7 are these four branches; §2 is the argument without pictures.

## 2 The argument in one page

**Signs and counts.** For the geodesic flow of a 3-manifold ( $q = \dim E_s = 2$ ) a resonance of  $L_X$  on  $k$ -forms is a zero of  $\zeta$  when  $k$  is even and a pole when  $k$  is odd. Write  $c_k(\lambda_0)$  for the algebraic multiplicity of  $\lambda_0$  as a degree- $k$  resonance: the dimension of the generalized eigenspace, equivalently the rank of the Riesz projector for a small disc around  $\lambda_0$  in degree  $k$ . It is unsigned, and a Jordan block counts by its size. The order of  $\zeta$  at  $\lambda_0$  is the alternating sum

$$\text{ord}_{\lambda_0} \zeta = \sum_k (-1)^k c_k(\lambda_0) = (\text{zeros at } \lambda_0) - (\text{poles at } \lambda_0),$$

positive for a zero, negative for a pole, 0 for a regular nonvanishing point.

**At the hyperbolic metric.** For the character  $\chi_\theta$ ,  $\theta \neq 0$  small, the twisted Laplacian on functions has a bottom eigenvalue  $\mu_0(\theta) = \Theta(\theta^2)$ , and the Dyatlov–Faure–Guillarmou correspondence turns it into a real scalar resonance

$$s^*(\theta) = -1 + \sqrt{1 - \mu_0(\theta)} \approx -\frac{1}{2}\mu_0(\theta) < 0, \quad s^*(\theta) \rightarrow 0 \text{ as } \theta \rightarrow 0$$

(§3): the *double point*, the place where everything below starts. The cluster at  $s^*(\theta)$  is  $c = (1, 2, 3, 2, 1)$  (§4): one scalar state  $f$ ; two degree-1 states, namely  $d_0 f$ , the lift of  $f$ , which is the *closed branch* (locked to the scalar resonance, it goes wherever  $f$  goes), and  $I d_0 f$ , its rotation by a quarter turn in  $E_u \oplus E_s$ , which is the *twin* (it exists only because the hyperbolic metric is conformal on  $E_u$ , and is tied to nothing); and their lifts and duals. Its order is  $1 - 2 + 3 - 2 + 1 = +1$ , Fried's simple zero. Each of the two degree-1 states is a pole on its own; the zero is the net of the whole cluster. The degree-1 cluster has rank two, which is why the Lean statement encodes it by a  $2 \times 2$  matrix  $M$ .

**Turning  $\tau$ .** The conformal deformation splits the double point. The scalar resonance  $s^*$  drifts only at rate  $O(\theta^2)$  and never reaches 0: a degree-0 resonant state at 0 would be a flow-invariant twisted function, a class in  $H^0(Z; \chi_\theta) = 0$ , so  $c_0 = 0$  at every metric of the family (DGRS Lemma 7.4). The closed branch, locked to it, stays near  $s^*$  as well. The twin leaves at rate  $r_0 > 0$ , dragging its lifts with it:  $d_0 u_{\text{nc}}$  in degree 2 (a zero) and  $d\alpha \wedge u_{\text{nc}}$  in degree 3 (a pole), locked to it because  $d_0$  commutes with the flow. The travelling chain is  $(0, 1, 1, 1, 0)$ , of order  $-1 + 1 - 1 = -1$ : a simple pole. Its position  $s_{\text{nc}}(\theta, \tau)$  is real (the resonance set is symmetric under  $s \mapsto \bar{s}$ ) and increases from  $s^*(\theta) < 0$ , so by the intermediate value theorem it vanishes at a unique  $\tau = \sigma(\theta)$ . At  $g_\sigma$  the number 0 is a genuine degree-1 resonance.

**Exactness forces order 0, not +1.** At the crossing, acyclicity gives  $c_0 = c_4 = 0$  (DGRS Lemma 7.4), Hodge duality gives  $c_3 = c_1$ , and exactness of the reduced complex  $C_0^1 \rightarrow C_0^2 \rightarrow C_0^3$  (Dang–Rivière) gives  $c_2 = c_1 + c_3 = 2c_1$ : the middle space is the image of the first map plus what maps onto the third. So  $\sum_k (-1)^k c_k = -c_1 + 2c_1 - c_1 = 0$  whenever the complex at 0 is exact. Order +1 is what happens at  $s^*$  at  $g_{\text{hyp}}$ , where the degree-0 state  $f$  is also present; at the crossing there is no degree-0 state at 0, so the order is 0:  $\zeta(\cdot; g_\sigma)$  is holomorphic and nonvanishing at  $\lambda = 0$ , and  $\zeta(0; g_\sigma)$  is an ordinary value of the meromorphic continuation, exactly as in Fried's identity itself.

**Who supplies the second degree-2 state.** The twin is the only degree-1 state at 0, so  $c_1 = 1$  and exactness demands  $c_2 = 2$ . Degree 2 near 0 holds four branches (the  $4 \times 4$  matrix  $A$ ): the lift of the closed branch, which sits near  $s^* \neq 0$  and is unavailable; the lift of the twin, which supplies one; and the two *pure* roots of  $Q = X^2 - e_1 X + e_2$ . The second state at 0 must therefore be a pure root, so  $e_2(\theta, \sigma) = 0$ ; and since the multiplicity is exactly 2, only one pure root is at 0, so  $e_1(\theta, \sigma) \neq 0$ . Call the arriving pure root  $z$ . Its partner, the spectator, sits at  $e_1(\theta, \sigma)$ , off 0 because  $\theta \neq 0$  (at  $\theta = 0$  both pure roots are pinned at 0 for every  $\tau$ ). Nothing ties  $z$  to the twin along the way: they coincide at the endpoint because exactness demands it, and nowhere else.

**The value.** Near the crossing Chaubet–Dang's factorisation, read at  $\lambda = 0$ , gives  $\zeta(\lambda; g_\tau) = F(\lambda, \tau) (\lambda - z(\tau)) / (\lambda - s_{\text{nc}}(\tau))$  with  $F$  holomorphic, nonvanishing and continuous in  $\tau$ : the finite cluster factor split off from the rest of the resolvent. At  $\tau = \sigma$  the factor is  $\lambda/\lambda = 1$ , so  $\zeta(0; g_\sigma) = F(0, \sigma)$ . Off the crossing  $\zeta(0; g_\tau) = \tau_R$  (DGRS Thm 2 plus Fried at  $g_{\text{hyp}}$ ), hence  $F(0, \tau) = \tau_R s_{\text{nc}}(\tau) / z(\tau)$ . Both branches cross 0 linearly in  $\tau$ , so continuity of  $F$  gives

$$\zeta(0; g_\sigma) = \tau_R \cdot \frac{s'_{\text{nc}}(\sigma)}{z'(\sigma)} = \tau_R \cdot \frac{\text{pole rate}}{\text{zero rate}}.$$

The only limit in the argument is this one, and it is a limit of the continuous factor  $F$ . The pole's rate is  $r_0$  up to small corrections. The zero's rate is  $\partial_\tau e_2/e_1$ , because the two pure roots multiply to  $e_2$  and the spectator is parked at  $e_1$ : the closer the spectator sits to 0, the steeper  $z$  must arrive. The Lean statement certifies that the twin's rate is at most  $2r_0$  and the zero's slope exceeds  $2r_0$  in modulus, so the ratio is not 1; informally it is  $\Theta(\theta^2)$ . Fried's identity would need the two speeds to be equal.

**What survives.** The ratio is the torsion of the zero cluster (Lemma A), so Chaubet–Dang's refined identity, cluster torsion included, holds at the crossing; the naive identity of §1 does not. Sections §3–§7 draw each step.

### 3 The dial $\theta$ : a character near trivial makes a small eigenvalue

Take  $Z$  a closed hyperbolic 3-manifold with  $b_1(Z) = 1$ ,  $\omega$  the harmonic 1-form generating  $H^1$ , and the family of unitary characters  $\chi_\theta = \exp(i\theta \int \omega)$ . For small  $\theta \neq 0$  the character is acyclic (the Alexander polynomial has finitely many roots on the unit circle), so  $\tau_R(\chi_\theta)$  is defined. The price of being near trivial is a small eigenvalue: the twisted Laplacian on functions has  $\Delta_\theta = \Delta + \theta A + \theta^2 |\omega|^2$  with  $A \cdot 1 = 0$ , so its bottom eigenvalue is

$$\mu_0(\theta) = \theta^2 \frac{\|\omega\|_{L^2}^2}{\text{vol}(Z)} + O(\theta^4), \quad \mu_0(\theta) \rightarrow 0 \text{ as } \theta \rightarrow 0.$$

The Dyatlov–Faure–Guillarmou correspondence turns Laplace eigenvalues into scalar (degree-0) resonances: with the convention  $(L_X + s)u = 0$ , each  $\mu_j$  produces the pair  $s = -1 \pm \sqrt{1 - \mu_j}$ . For  $\mu_0$  both roots are real and simple:

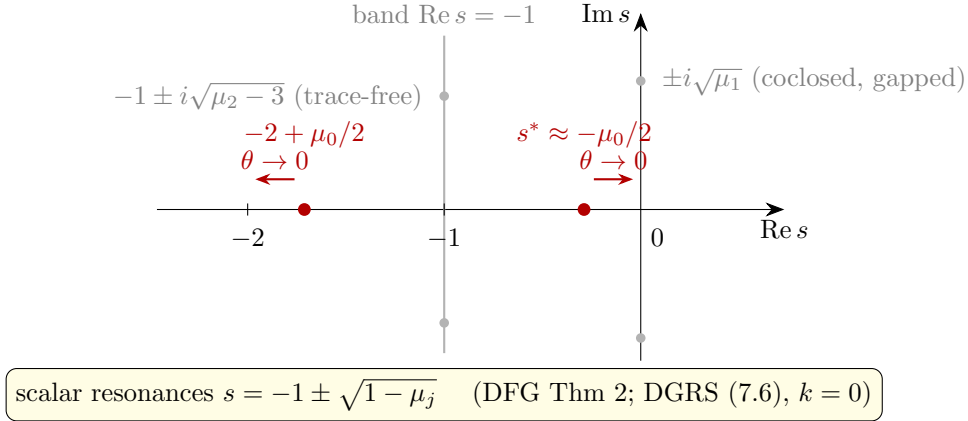
$$s^*(\theta) = -1 + \sqrt{1 - \mu_0} = -\frac{\mu_0}{2} - \frac{\mu_0^2}{8} - \dots \approx -\frac{\mu_0(\theta)}{2}, \quad -1 - \sqrt{1 - \mu_0} \approx -2 + \frac{\mu_0}{2}.$$

As  $\theta \rightarrow 0$  the first slides toward 0 and the second toward  $-2$ . Everything else in the spectrum near the real axis stays a fixed distance away: the trace-free sector lives on the band  $\text{Re } s = -1$  (a Bochner margin), and the coclosed 1-form sector sits at  $\pm i\sqrt{\mu_1}$  with  $\mu_1$  bounded below uniformly in  $\theta$ , because the one small eigenform of the twisted Hodge Laplacian on 1-forms is exact and cancels against functions. In the Lean statement this is the hypothesis **hdouble**:  $s^*(0) = 0$ ,  $s^*$  continuous at 0,  $s^*(\theta) < 0$  for  $\theta \neq 0$ .

### 4 The twin: every scalar resonance appears twice in degree 1

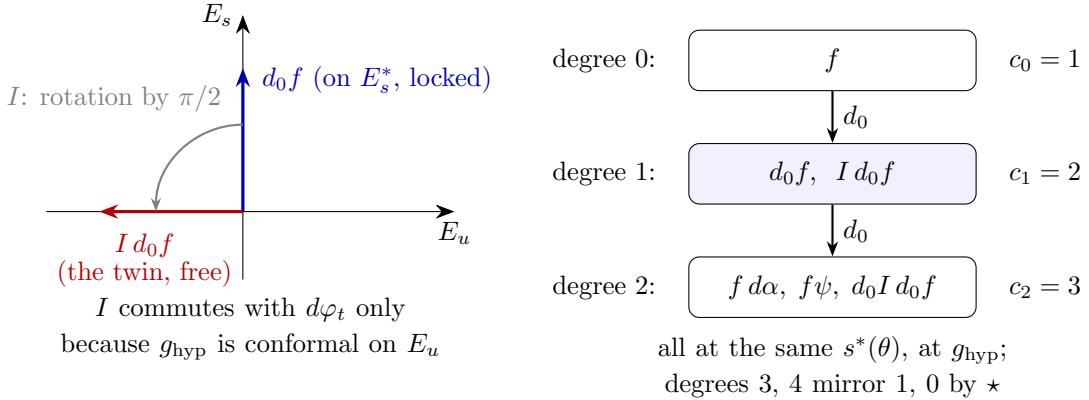
A scalar resonant state  $f$  at  $s^*$  lifts to degree 1 by the horizontal differential  $d_0 f := df - (Xf)\alpha = df + s^* f \alpha$ , which is annihilated by  $\iota_X$  and satisfies  $L_X d_0 f = d_0 L_X f = -s^* d_0 f$ . So  $s^*$  is a degree-1 resonance too. At  $\theta = 0$  this state is closed; at  $s^* \neq 0$  it is not ( $d(d_0 f) = s^*(d_0 f \wedge \alpha + f d\alpha)$ ), and the accurate description is that it is *locked to the degree-0 branch*: wherever the scalar resonance goes,  $d_0 f$  goes with it. We keep the name “closed branch” as a label.

The hyperbolic metric supplies a second state at the same  $s^*$ . Because  $d\varphi_t$  is conformal on  $E_u$  (and on  $E_s$ ), rotation by  $\pi/2$  on  $E_u \oplus E_s$  is a smooth tensor  $I$  that commutes with the flow, annihilates  $X$ , and acts trivially on the twisted bundle  $E$ . Hence  $I d_0 f$  is also resonant at  $s^*$ . It



**Figure 2:** The first dial. In the  $s$ -plane, the small twisted eigenvalue  $\mu_0(\theta)$  produces two simple *real* scalar resonances; the one at  $s^*(\theta) \approx -\mu_0/2$  walks toward 0 as  $\theta \rightarrow 0$ , its partner toward  $-2$ . Nothing else comes near the real axis: the trace-free sector is pinned to the band  $\operatorname{Re} s = -1$  and the coclosed sector to  $\pm i\sqrt{\mu_1}$ . Nothing crosses yet —  $s^* < 0$  for every  $\theta \neq 0$ .

is independent of  $d_0 f$ : first-band states are totally unstable,  $d_0 f$  lives on  $E_s^*$ , and  $I d_0 f \propto d_0 f$  would force the boundary datum on  $S^2$  to be constant, impossible for a nontrivial character. So in degree 1 the scalar resonance  $s^*$  has multiplicity exactly two, with states  $d_0 f$  and  $I d_0 f$ ; the cluster has rank two, so the double point is semisimple. The twin is not locked to anything: it exists only because the hyperbolic metric is conformal on  $E_u$ , and it will move as soon as that conformality is broken. In the Lean statement the two states are the hypothesis **hstates**, and semisimplicity is derived from them; the independence argument of this paragraph is the one input that is derived here rather than read from a cited paper.



**Figure 3:** The twin. Left: in the  $E_u \oplus E_s$  plane the locked state  $d_0 f$  and its rotation  $I d_0 f$  are independent resonant 1-forms at the same  $s^*$ . Right: the cluster at  $s^*$  across degrees,  $c = (1, 2, 3, 2, 1)$ , net order  $1 - 2 + 3 - 2 + 1 = +1$ : one simple zero of  $\zeta$  at  $s^*$ , as Fried's factorization at the hyperbolic metric requires ( $\zeta(0; g_{\text{hyp}}) = \tau_R$ , finite and nonzero). The partner  $-s^*$  carries a second simple zero (state  $f'\omega_s$ ). In the Lean statement the degree-2 spectrum  $\{s^*, s^*, s^*, -s^*\}$  at  $\tau = 0$  is the hypothesis **hmirror**.

## 5 The second dial $\tau$ : a conformal deformation unlocks the twin

Now deform the metric conformally,  $g_\tau = e^{-2\tau b} g_{\text{hyp}}$ , with  $b$  in the open dense set of functions for which CDDP's first variation is nondegenerate. (CDDP write the deformation parameter as  $s$ ; we write  $\tau$  to keep  $s$  for the spectral variable.) Pulling the new geodesic flow back to the fixed phase space, the double point at  $s^*$  splits. The two velocities are read off CDDP's first-variation formula, equations (4.22) and (4.38): the locked state moves with the degree-0 branch, at rate  $O(\theta^2)$ ; the twin moves at rate

$$r_0 = -\frac{\int (b \circ \pi) \alpha \wedge du_\psi \wedge du_\psi^*}{\langle\langle u_\psi, d\alpha \wedge u_\psi^* \rangle\rangle} \neq 0,$$

real, with  $O(\theta^2)$  corrections. Write  $s_{\text{nc}}(\theta, \tau)$  for the twin's position, so  $s_{\text{nc}}(\theta, 0) = s^*(\theta) \approx -\mu_0/2$  and  $\partial_\tau s_{\text{nc}}(0, 0) = r_0$ . In the Lean statement the first-variation formula is the hypothesis **h422** (the pairing matrix  $B$  and  $B \cdot \partial_\tau M(0, 0) = \text{diag}(0, p)$ ) and  $r_0 = B_{cc} p / \det B > 0$  is **hr<sub>0</sub>**; the twin is the larger root  $s^* + \tau w_+$  of the characteristic polynomial of  $M$ .

Two facts make the walk to zero unavoidable rather than accidental. *Reality*: the twisted resonance set is symmetric under  $s \mapsto \bar{s}$  for every unitary  $\rho$  (Hermitian adjoint composed with time reversal  $J(x, v) = (x, -v)$  preserves  $\rho$ ; complex conjugation sends  $\rho$  to  $\bar{\rho}$  and, for the character family,  $\theta$  to  $-\theta$ ). A simple resonance near the real axis is therefore real, and stays real along a real one-parameter family. *No mirror*: there is no  $s \mapsto -\bar{s}$  symmetry off  $g_{\text{hyp}}$ , so nothing stops a real resonance from passing through 0. Hence  $\{g : 0 \in \text{Res}\}$  is locally the zero set of one real-analytic function: crossing has **codimension one**. The implicit function theorem at  $(\theta, \tau) = (0, 0)$  gives the crossing curve

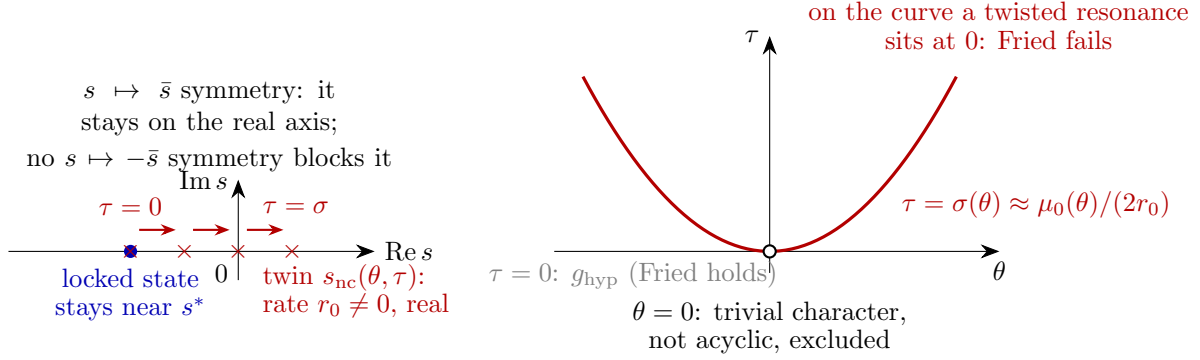
$$\sigma(\theta) = \frac{\mu_0(\theta)}{2r_0} + O(\theta^4) = \frac{\theta^2 \|\omega\|^2}{2r_0 \text{vol}(Z)} + O(\theta^4), \quad s_{\text{nc}}(\theta, \sigma(\theta)) \equiv 0.$$

At  $(g_{\sigma(\theta)}, \chi_\theta)$  the twin is a nonzero, non-closed, twisted resonant 1-form at  $s = 0$ ; the character is still acyclic (cohomology does not see the metric) and the metric is still negatively curved with  $|K + 1| = O(\theta^2)$ . The Lean statement proves existence and uniqueness of the crossing in  $(0, 2|s^*|/r_0]$  from **hdouble**, **h422**, **hr<sub>0</sub>** and the regularity of  $M$ ; the parabola above is the informal sharpening.

## 6 Zero and pole arrive at different speeds

What does the crossing do to  $\zeta$ ? The exponent rule is  $\zeta = \prod_k Z_{X(k)}^{(-1)^k}$ : degree-0, 2, 4 resonances are zeros, degree-1, 3 resonances are poles. Follow the twin  $u_{\text{nc}}$  off the hyperbolic metric. It drags a chain with it:  $d_0 u_{\text{nc}} \neq 0$  in degree 2 and  $Lu_{\text{nc}} = d\alpha \wedge u_{\text{nc}}$  in degree 3 are resonant at the same  $s_{\text{nc}}$ , a  $(0, 1, 1, 1, 0)$  cluster of net order  $-1 + 1 - 1 = -1$ : **a pole of  $\zeta$** , heading for 0 at rate  $r_0$ . In the Lean statement the lift  $d_0 u_{\text{nc}}$  is the hypothesis **hlocked**: every eigenvalue of  $M$  is an eigenvalue of  $A$ .

Meanwhile the degree-2 cluster has rank 4. Two of its branches are the locked branches (the lifts of the closed branch and of the twin); the other two are the *pure* degree-2 resonances  $\beta_3, \beta_4$ , the roots of  $Q = X^2 - e_1 X + e_2$ , each a simple zero of  $\zeta$ . At  $g_{\text{hyp}}$  they sit at  $\pm s^*$  (states  $f\psi$  and  $f'\omega_s$ , the second from the stable area form, which lands exactly at  $-s^*$ ). At



**Figure 4:** The second dial. Left: as  $\tau$  grows the twin walks along the real axis at rate  $r_0$  and passes through 0 at  $\tau = \sigma(\theta)$ ; the locked state stays with the scalar branch. Right: in the  $(\theta, \tau)$  plane the crossing locus is a real-analytic curve through the origin, tangent to the  $\theta$ -axis, even in  $\theta$ ; drawn with  $r_0 > 0$ . The  $\theta$ -axis is the hyperbolic metric. The origin itself is the trivial character and is excluded. On the curve a twisted resonance sits at 0; off it none does.

$\theta = 0$  they are pinned at 0 for every  $\tau$  (CDDP Cor 4.1:  $m_{2,0}(0) = b_1 + 2$ , so the pole leaves 0 *alone*; in Lean, **hpinning**). Two consequences follow for their sum  $e_1 = \beta_3 + \beta_4$  and product  $e_2 = \beta_3\beta_4$ . First,  $e_1$  vanishes on both axes, so  $e_1(\theta, \sigma(\theta))$  is small:  $O(\theta^4)$  using evenness in  $\theta$ , and  $|e_1(\theta, \sigma)| \leq K_1|\theta|\sigma$  is what the Lean statement certifies from  $C^2$  regularity alone. Second, at the crossing the resonant complex must still be exact (Dang–Rivière; in Lean, **hexact**), and exactness with  $c_1 = c_3 = 1$ ,  $c_0 = c_4 = 0$  forces  $c_2 = 2c_1 = 2$ : a pure degree-2 zero must arrive at 0 at the very instant the pole does, i.e.  $e_2(\theta, \sigma) = 0$ . So the crossing cluster is  $c = (0, 1, 2, 1, 0)$ , of order  $-1 + 2 - 1 = 0$ :  $\zeta$  is regular and nonzero at 0. The pole has been cancelled by a zero that is a *different* resonance.

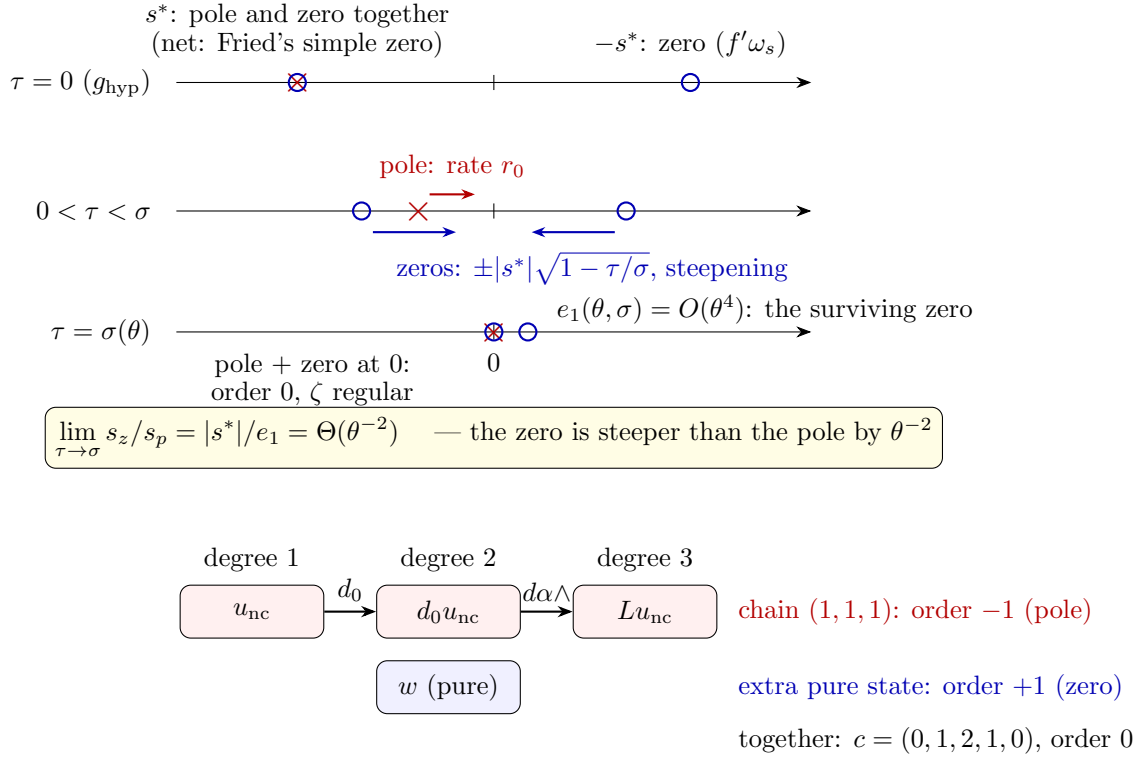
Cancelled, but not balanced. Near  $\tau = \sigma$  the pole sits at  $s_p \sim r_0(\tau - \sigma)$  and the arriving zero at  $s_z = z(\theta, \tau)$ , the pure branch nearest 0, and

$$\lim_{\tau \rightarrow \sigma} \frac{s_z}{s_p} = \frac{|s^*(\theta)|}{e_1(\theta, \sigma(\theta))} (1 + O(\theta^2)) = \frac{\Theta(\theta^2)}{O(\theta^4)} = \Theta(\theta^{-2}) \rightarrow \infty.$$

The zero comes in from farther and steeper: its slope at the crossing is  $\Theta(\theta^{-2})r_0$ , against the pole's  $r_0$ . Lemma A (the spectral-flow formula for the refined torsion of the zero cluster:  $(-1)^{Q_0\tau}(C^\bullet(0), \Gamma_\vartheta) = \lim \prod (-s_0(\tau))^{(-1)^{q_m(s_0(\tau))}}$  over the resonances converging to 0) then reads  $\tau(C^\bullet(0)) = -\lim s_z/s_p$ , and Cauchy's formula on a fixed circle (Chaubet–Dang (6.5) read at  $\lambda = 0$ ; in Lean, **hfactor**) gives the value at the crossing:

$$\zeta_{\chi_\theta}(0; g_{\sigma(\theta)}) = \tau_R \cdot \lim_{\tau \rightarrow \sigma} \frac{s_p}{s_z} = \tau_R \cdot \frac{e_1(\theta, \sigma)}{|s^*(\theta)|} = \Theta(\theta^2) \tau_R \neq \tau_R.$$

The Lean statement certifies the weaker, sufficient form: the twin's rate at the crossing is at most  $2r_0$ , the zero's slope  $\partial_\tau e_2/e_1$  exceeds  $2r_0$  in modulus, so the ratio is not 1; the  $\Theta(\theta^{\pm 2})$  sizes are the informal sharpening.



**Figure 5:** Different speeds. Top: three snapshots of the real axis near 0. At  $g_{\text{hyp}}$  the pole ( $\times$ ) and one pure zero ( $\circ$ ) coincide at  $s^*$  and net to Fried's simple zero; as  $\tau$  grows the pole moves at rate  $r_0$  while the two pure zeros close in from  $\pm|s^*|\sqrt{1-\tau/\sigma}$  along a square root, steepening; at  $\tau = \sigma$  the pole and one zero meet at 0 and the other zero rests at  $e_1 = O(\theta^4)$ . Bottom: the degree ladder at the crossing. The chain through  $d_0$  and  $d\alpha\wedge$  is the pole; exactness forces one more pure degree-2 state, and  $c_2 = 2c_1$  makes the order zero. Regularity is exact; balance is not.

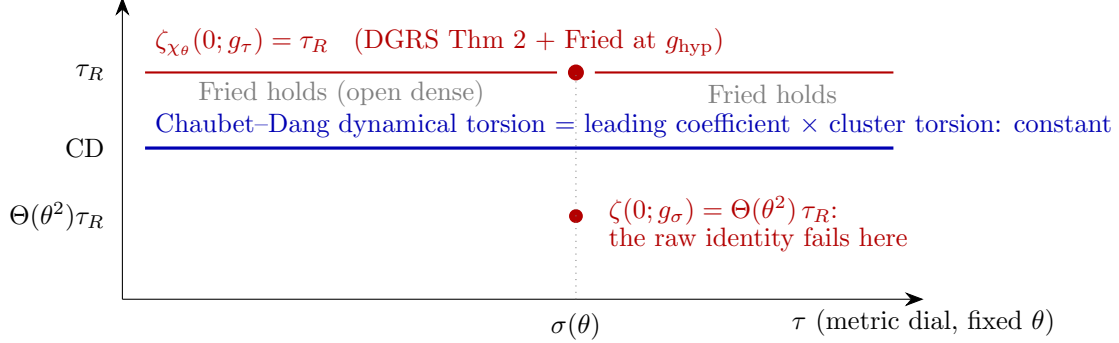
## 7 The jump, and what survives

Assemble the picture along the deformation. For  $\tau \neq \sigma(\theta)$  no resonance is at 0, and DGRS Theorem 2 (local constancy of  $\zeta(0)$  on the open set where 0 is not a resonance) together with Fried at  $g_{\text{hyp}}$  gives  $\zeta_{\chi_\theta}(0; g_\tau) = \tau_R$  (in Lean, `hfried.off`). At  $\tau = \sigma(\theta)$  the value is  $\Theta(\theta^2)\tau_R$ . So the function  $g \mapsto \zeta_g(0)$  is constant on the hyperbolic component minus the crossing hypersurface and takes a different value on it: a removable-looking discontinuity that is not removable, because the value at the point is what Fried's identity is about. The factor  $\theta^2$  is an arrival ratio, not a property of the torsion:  $\tau_R(\chi_\theta)$  is itself of size  $\theta^4$  at fixed  $\theta$ , and the jump is measured relative to it.

Chaubet–Dang's refinement reads the same picture without a jump. Their Theorem 4 says that along any path of negatively curved metrics from  $g_{\text{hyp}}$ ,  $\zeta(s)^{(-1)^q} = |s|^{(-1)^q m} \tau_R \cdot \tau(C^\bullet(X_{g_0}))/\tau(C^\bullet(X_g))(1+O(s))$ , where  $\tau(C^\bullet(X_g))$  is the refined torsion of the finite-dimensional zero-resonant complex; Corollary 8 packages the leading Laurent coefficient times the cluster torsion into a constant on connected open sets. On the crossing curve  $m = 0$ , the leading coefficient is  $\zeta(0) = \Theta(\theta^2)\tau_R$ , and the cluster torsion is  $|\tau(C^\bullet(0))| = |s^*|/e_1 = \Theta(\theta^{-2})$ ; the product is  $\tau_R$ , continuous across the hypersurface. Off the curve the cluster is empty, its torsion is 1, and the



two readings agree. The dynamical torsion is the invariant that does not jump; the raw value  $\zeta(0)$  is not. Readers who reserve the name “Fried’s conjecture” for the resonance-free case will read the construction as an instance of the refined identity rather than as a counterexample; the identity that fails is the naive one displayed in §1.



**Figure 6:** The jump. A one-dimensional slice of metric space at fixed  $\theta \neq 0$ . The raw value  $\zeta(0)$  (red) is flat at  $\tau_R$  except at the single crossing point, where it drops to  $\Theta(\theta^2)\tau_R$ : a value discontinuity with no order change. The dynamical torsion (blue) is flat throughout, because the cluster torsion  $\Theta(\theta^{-2})$  compensates exactly. Fried at generic metrics of the hyperbolic component is a theorem (Humbert–Tao Thm 5 + DGRS Prop 7.7 + Chaubet–Dang Thm 4); at the crossing metrics the raw identity fails and only the refined one survives.

What survives, then, is the refined statement: the correct invariant at  $s = 0$  is the dynamical torsion, and Fried’s conjecture as usually written needs the cluster factor at the codimension-one set of metrics where a twisted resonance sits at 0.

## 8 Status

**What is machine-checked.** The Lean 4 / Mathlib theorem `fried_counterexample_of_resolvent_inputs`, in the namespace `FriedCrossing`, is a conditional reduction. Its statement of record is `Challenge.lean`, generated from the statement-definitions module `B1s/fried_statement_defs_9_14.lean`; its proof is in `B1s/fried_counterexample_main.lean` and the leg files that module imports. Its objects are two abstract resolvent families on Banach scales, complex structures and nondegenerate frames, from which the cluster matrices  $M$  and  $A$  of §1 are *defined* by contour integrals; its hypotheses are, in six groups, the properties of that data that the argument uses, each cited to the literature: the two states at the double point (`hstates`, §4); the first variation and  $r_0 > 0$  (`h422`, `hr0`; CDDP (4.22), (1.3)); the double point itself (`hdouble`; DGRS (7.6), Bunke–Olbrich, DFG); the mirror spectrum at  $\tau = 0$  (`hmirror`; DGRS (7.6)); the pinning at  $\theta = 0$  (`hpinning`; CDDP Cor 4.1); the locked branches (`hlocked`; DGRS Lemma 7.1); and, per  $\theta$ , the  $\zeta$ -side facts (`hfactor`, Chaubet–Dang (6.5); `hfried_off`, DGRS Thm 2 with Fried at  $g_{hyp}$ ; `hexact`, Dang–Rivière; the dimension bookkeeping `hacyc`, `hdual`, `hdims`). Its conclusion is that the twin crosses 0 at exactly one  $\sigma \in (0, 2|s^*|/r_0]$ , that  $\zeta$  has order 0 there, that the zero’s slope exceeds the pole’s, and that  $\zeta(0; g_\sigma) = \tau_R \cdot (\text{pole rate})/(\text{zero rate}) \neq \tau_R$ , with the Lemma A torsion identity. The theorem depends on no axiom beyond Lean’s three standard ones, and `leanprover/comparator` matches it constant by constant against the statement of record.

**What is not.** The identification of the abstract data with the geometry — that the resolvent families are those of the twisted deformed generator on the anisotropic spaces of  $g_\tau$ , that  $\zeta_0(\tau)$  is the Ruelle zeta function of  $g_\tau$  at 0, that  $\tau_R$  is the Reidemeister torsion of  $\chi_\theta$ , and that the defined matrices are the cluster matrices (their imaginary parts vanish) — is the informal layer of this document and the accompanying note. Nothing about manifolds, flows, anisotropic spaces or zeta functions is in Lean, and Lean does not assert that data satisfying all the hypotheses at once exist. Of the hypotheses, only the independence of  $d_0f$  and  $I d_0f$  inside `hstates` is derived (in §4) rather than read from a cited paper; the frame nondegeneracy is a choice. The  $\Theta(\theta^{\pm 2})$  sizes drawn in §6–§7 are informal sharpenings of the certified inequalities.

**Caveat.** This is a conditional reduction from cited inputs. It has not yet been verified by a mathematician other than the author, and the claim that Fried’s identity fails at the crossing metric stands or falls with the six groups of cited facts holding as stated for the geodesic flow.

**Disclosure.** This document, the accompanying note and the Lean development were produced with substantial assistance from Claude language models directed by the author, who takes responsibility for the mathematics. The model roles are recorded in the repository’s metadata file `formalization.yaml`.